Description of Additional Supplementary Files

Supplementary Data 1. Comparison of all features across different subphenotypes derived from the development cohort. P-value: The p value for testing the differences of each variable across the subphenotypes. Continuous variables were tested by one-way ANOVA (normal) or Kruskal–Wallis test (non-normal), and categorical variables were tested by one-sided Fisher's exact test. Post hoc pairwise analysis: Continuous variables were tested by Tukey's honestly significant difference (normal) or pairwise Wilcoxon rank sum test (non-normal), and categorical variables were tested by pairwise Fisher's test for each pair of subphenotypes to identify pairwise significant differences if the overall p-value for three groups was < 0.05. For instance, 1 vs. 2 means that subphenotypes 1 and 2 have significant differences on the variable with p-value < 0.05.

Supplementary Data 2. Comparison of all features across different subphenotypes derived from the validation cohort. P-value: The p value for testing the differences of each variable across the subphenotypes. Continuous variables were tested by one-way ANOVA (normal) or Kruskal–Wallis test (non-normal), and categorical variables were tested by one-sided Fisher's exact test. Post hoc pairwise analysis: Continuous variables were tested by Tukey's honestly significant difference (normal) or pairwise Wilcoxon rank sum test (non-normal), and categorical variables were tested by pairwise Fisher's test for each pair of subphenotypes to identify pairwise significant differences if the overall p-value for three groups was < 0.05. For instance, 1 vs. 2 means that subphenotypes 1 and 2 have significant differences on the variable with p-value < 0.05.

Supplementary Data 3. The list of all included features.

Supplementary Data 4. The comorbidity classes and their corresponding ICD-9/ICD-10 codes.

Supplementary Data 5. The lab test features and their corresponding LONIC Codes.